

AL JOUF, Saudi Arabia

WMO: 403610

Lat: 29.785N

Lon: 40.100E

Elev: 689

StdP: 93.31

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	3.2	-15.1	1.1	10.9	-12.8	1.3	13.2	13.1	11.5	11.4	13.1	2.2	240	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.7	44.0	19.0	42.9	18.8	41.7	18.4	20.2	40.3	19.5	39.9	18.8	39.1	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.9	21.6	13.0	10.1	21.5	11.9	9.4	22.7	61.2	40.5	58.4	39.7	56.1	39.1	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.6	8.5	DB	-1.7	45.7	2.0	0.7	-3.1	46.3	-4.3	46.7	-5.5	47.1	-6.9	47.7	(4)
(5)			WB	-4.3	21.1	2.0	1.1	-5.8	21.9	-7.0	22.5	-8.1	23.1	-9.6	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	11.0	13.5	17.6	22.7	28.0	31.8	33.5	33.9	31.1	25.7	17.6	12.5	(6)	
	DBStd	8.78	3.15	3.46	3.94	3.64	3.20	2.34	2.49	2.33	2.57	3.26	3.77	3.26	(7)	
	HDD10.0	44	24	7	1	0	0	0	0	0	0	0	1	12	(8)	
	HDD18.3	675	227	138	62	6	0	0	0	0	0	1	58	183	(9)	
	CDD10.0	4900	55	106	237	381	558	654	728	742	632	488	229	89	(10)	
	CDD18.3	2491	1	4	40	137	301	404	470	483	382	231	36	2	(11)	
	CDH23.3	37014	12	66	473	1562	4102	6357	7789	8007	5694	2618	308	26	(12)	
(13)	CDH26.7	23631	1	13	169	706	2394	4204	5419	5622	3662	1354	84	2	(13)	
(14)	Wind	WSAvg	3.8	3.5	3.9	4.3	4.3	4.2	4.3	3.9	3.4	3.3	3.2	3.1	(14)	
(15) Precipitation	PrecAvg	51	12	7	5	3	2	0	0	0	1	4	8	9	(15)	
	PrecMax	96	33	27	26	14	12	3	0	2	14	28	61	41	(16)	
	PrecMin	10	1	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	24	10	8	7	4	3	0	0	0	3	7	14	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	28.8	34.8	37.1	41.1	44.0	45.2	45.2	43.8	38.3	31.8	26.2	(19)	
		MCWB	12.3	13.5	14.5	16.0	17.8	18.3	19.5	19.4	18.8	17.2	14.8	12.6	(20)	
	2%	DB	21.8	25.1	30.9	34.9	39.1	42.1	44.0	44.1	41.8	36.8	29.1	23.1	(21)	
		MCWB	10.4	11.4	13.8	15.6	17.0	18.1	19.0	19.2	18.4	16.5	14.1	11.7	(22)	
	5%	DB	19.1	22.9	28.1	33.0	37.8	40.9	42.8	42.9	40.1	35.1	27.0	20.8	(23)	
		MCWB	10.0	10.9	12.7	14.8	16.6	17.8	18.7	19.1	17.9	16.1	13.5	10.8	(24)	
	10%	DB	17.1	20.8	25.9	30.9	36.0	39.2	41.1	41.5	38.8	33.2	24.8	18.8	(25)	
		MCWB	9.1	10.0	11.8	14.0	16.0	17.2	18.2	18.7	17.6	15.7	12.9	10.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	14.7	16.2	17.7	19.2	20.1	21.1	21.5	20.6	18.7	17.2	15.0	(27)	
		MCDB	19.1	23.2	27.1	29.5	36.0	40.4	41.5	41.0	39.0	33.4	24.1	20.5	(28)	
	2%	WB	12.5	13.0	14.8	16.6	17.9	19.1	19.9	20.5	19.0	17.6	16.0	13.3	(29)	
		MCDB	18.7	21.5	26.8	30.3	34.9	39.7	40.9	40.4	38.1	31.8	22.3	20.2	(30)	
	5%	WB	10.9	11.8	13.6	15.7	17.2	18.2	19.1	19.7	18.4	16.9	15.0	12.0	(31)	
		MCDB	17.7	19.9	26.2	29.6	34.9	39.1	40.0	40.4	37.6	31.2	22.7	18.9	(32)	
	10%	WB	9.6	10.9	12.4	14.8	16.5	17.5	18.5	18.9	17.8	16.3	14.1	10.9	(33)	
		MCDB	16.2	19.1	24.4	28.8	34.1	38.2	39.2	40.1	37.0	30.5	22.6	17.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.3	13.1	13.6	13.7	14.4	14.3	14.7	14.6	13.6	12.0	11.7	(35)	
		MCDBR	14.0	15.3	15.9	15.5	15.2	15.5	15.1	15.2	15.5	15.0	14.2	14.0	(36)	
	5% WB	MCWBR	7.0	7.2	7.1	6.4	5.8	5.8	5.5	5.7	5.8	5.9	6.1	7.0	(37)	
		MCWBR	12.3	13.0	14.3	14.0	14.1	14.9	14.5	14.8	14.7	13.4	11.4	11.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.344	0.380	0.434	0.516	0.520	0.456	0.441	0.463	0.438	0.462	0.402	0.362	(40)	
	taud		2.186	2.062	1.907	1.734	1.746	1.909	1.979	1.922	1.962	1.892	2.023	2.125	(41)	
	Ebn at Noon		875	872	847	791	788	837	848	826	833	774	796	828	(42)	
	Edh at Noon		122	151	189	233	231	195	182	190	176	177	143	124	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.85	4.84	6.00	6.88	7.58	8.28	8.00	7.38	6.45	5.13	4.03	3.55	(44)	
	RadStd		0.19	0.23	0.20	0.22	0.32	0.15	0.21	0.17	0.18	0.19	0.28	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.39	N/A	N/A	+1.10	N/A	N/A	-32	-143	+462	+306	(46)
(47)	Regional (1 neighbor)	+1.39	N/A	N/A	+1.10	N/A	N/A	-32	-143	+462	+306	(47)

Nomenclature: See separate page